Microtubule sliding activity of a kinesin-8 promotes spindle assembly and spindle length control

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Evidence abstract

Molecular motors play critical roles in the formation of mitotic spindles, either through controlling the stability of individual microtubules, or by crosslinking and sliding microtubule arrays.

Kinesin-8 motors are best known for their regulatory roles in controlling microtubule dynamics.

They contain microtubule-destabilizing activities, and restrict spindle length in a wide variety of cell types and organisms.

Here, we report an antiparallel microtubule-sliding activity of the budding yeast kinesin-8, Kip3.

The in vivo importance of this sliding activity was established through the identification of complementary Kip3 mutants that separate the sliding activity and microtubule-destabilizing activity.

In conjunction with Cin8, a kinesin-5 family member, the sliding activity of Kip3 promotes bipolar spindle assembly and the maintenance of genome stability.

We propose a slide-disassemble model where the sliding and destabilizing activity of Kip3 balance during pre-anaphase.

This facilitates normal spindle assembly.

However, the destabilizing activity of Kip3 dominates in late anaphase, inhibiting spindle elongation and ultimately promoting spindle disassembly.

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